INCH-POUND

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PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8932 *

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: UHF, beam power tetrode, pulse amplifier, transmitting.
See figure 1.
Mounting position: Tube axis vertical, either end up.
Weight: 38 pounds nominal.

ABSOLUTE RATINGS:
(Anode pulsed amplifier, class B. 4/)

Parameter:	Ef	ebb	Ebb	ec2	Ec1	ib	Ib	ic1	Ic1	ic2
Unit:	Vac 1/	kv 2/	kV dc	kv 3/	V dc	a	mA dc	a 5/	mA dc	a
On time: 15 µs										
Maximum	1.05 (dc) 1.05 (60 Hz) 1.05 (400 Hz)	45	---	2.2	-400	80	320	15	60	15
Minimum	---	---	---	---	---	---	---	---	---	---
On time: 60 µs										
Maximum	1.05 (dc) 1.05 (60 Hz) 1.05 (400 Hz)	30	20	2.2	-400	60	1,200	5	90	5
Minimum	---	---	---	---	---	---	---	---	---	---
Test conditions:	---	---	---	---	---	---	---	---	---	---

ABOLUTE RATINGS:
(Anode pulsed amplifier, class B. 4/)

Parameter:	Ic2	Pi	Pp	tk	Cooling	Environmental requirements	F1	Fault protection	Duty	If (surge)
Unit:	mA dc	kW	kW	sec 6/	7/	9/	MHz	10/	---	A 6/
On time: 15 µs										
Maximum	60	16	14	---	---	---	550	---	0.004	550
Minimum	---	---	---	90	---	---	---	---	---	---
On time: 60 µs										
Maximum	90	16	14	---	---	---	550	---	0.02	550
Minimum	---	---	---	90	---	---	---	---	---	---
Test conditions:	---	---	---	---	8/	---	---	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification: Required.
Holding period: t = 96 hours

* Formerly identified as tube type 4666 (see 28/).

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Anode water pressure differential	1155	No voltages applied 13/	0.65	II	PPD	---	10.0	psi
Grid(1) water pressure differential	1155	No voltages applied 13/	0.65	II	GPD	---	6.0	psi
Grid(2) water pressure differential	1155	No voltages applied 13/	0.65	II	SPD	---	8.0	psi
Filament water pressure differential	1155	No voltages applied 13/	0.65	II	FPD	---	8.0	psi
Cathode water pressure differential	1155	No voltages applied 13/	0.65	II	CPD	---	8.0	psi
Electrode voltage (grid)	1261	Ebb = 15 kV dc; ec2 = 2 kv; Ec1/ib = 2.7 a; Ef = 0.95 V, 60 Hz; prr = 300 ± 5; tp = 60, +5, -0 µs	0.65	II	Ec1	---	-320	V dc
Filament current	1301	Ef = 0.95 V, 60 Hz	0.65	II	If	460	530	A
Peak power output	---	Ef = 0.95 V, 60 Hz; Ebb = 15 kV dc; ec2 = 1.8 kv; Ec1 = -250 ± 10 V dc; pd/ib = 60 a; prr = 300 ± 5; tp = 60 +5, -0 µs; F = 400 MHz (min) 1/ 16/ 18/ 26/ 27/ 30/	0.65	II	po Ic1 Ic2 Power gain	225 10 10 50	290 --- --- ---	kw mA dc mA dc ---
Emission check	---	Peak power output; Ef/If = 450A, 60 Hz; ec2 = 2 kv (max); ib = 60 a (max) 29/	0.65	II	po	225	---	kw
Perveance	---	Ef = 0.95 V, 60 Hz; ec1 = ec2 = 200 v; tp = 1.0 µs (min) prr = 10 (min) 1/ 17/	0.65	II	is	54	---	a
Electrode current (grid)	1256	Ef = 0.95 V, 60 Hz; Eb = 6.5 kV dc; Ec2 = 600 V dc; Ec1/Ib = 1.0 Adc; t = 5 minutes 19/ 25/	0.65	II	Ic1	---	-50	µA dc

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1 - Continued								
Electrode current (screen)	1256	Ef = 0.95 V, 60 Hz; Ec2 = 600 V dc; Ec1/Ib = 1.0 Adc; Eb = 6.5 kV dc	0.65	II	Ic2	-25	25	mA dc
RF aging time	---	Peak power output except, Ebb = 16.5 kV dc; Ib = 1.2 Adc; ec2 = 2.3 kv; Ec1 = -310 ± 40 V dc (Notes 18/ 26/ 27/ and 30/ do not apply) 14/	0.65	II	t	10	---	min
Input resonant frequency	---	20/	0.65	II	Fi	230	245	MHz
Output resonant frequency	---	20/	0.65	II	Fo	241	245	MHz
Conformance inspection, part 2								
Direct-interelectrode capacitance	1331	Use special shield adapter DSCC Drawing No. 66060, or equivalent 21/	---	---	Cg2k Cg1p	10,000 ---	18,000 0.15	pF pF
Conformance inspection, part 3								
Life test	---	Group S	---	---	---	---	---	---
Initial life	---	Ef = 0.95 V, 60 Hz; ebb = 30 kv; ec2 = 2 kv (max); Ec1 16/; pd/Ib = 165 mA dc; po = 750 kw (min); F = 225 ± 15 MHz; tp = 10 ± 1 µs; prr = 300 ± 5; specified life = 1,000 hours 1/ 11/ 12/ 15/ 16/ 31/	---	---	---	---	---	---
Initial life-test end point:	---							
Peak power output	---	Initial life	---	---	po	750	---	kw
Multiple short life (1)	---	Initial life except, po = 825 kw (min); t = 50 hours 32/	---	---	---	---	---	---

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3 - Continued								
Multiple short life-test end point (1):	---							
Peak power output	---	Multiple short life (1)	---	---	po	825	---	kw
Periodic-check tests								
Vibration	---	22/ 23/	---	---	---	---	---	---
Shock	---	22/ 24/	---	---	---	---	---	----
Vibration and shock test end point:	---							
Peak power output	---		---	---	po	225	---	kw

1/ Because the filament voltage, when operated near the maximum value, provides emission in excess of any requirements within tube rating, during life, the filament voltage shall be reduced to a value that will give adequate, but not excessive emission. Careful attention to maintaining the value consistent with adequate emission will result in conserving the life of the tube. The filament voltage shall be measured at the respective liquid coolant connections on the tube side of the threads. This procedure is essential in order to accurately measure the filament voltage. At 400 Hz some heating of the filament leads and RF cathode terminal (cathode header) occurs; this condition is not detrimental to tube operation or to tube life.

2/ Peak positive supply voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portion of the pulse. The magnitude of any spike, on the anode voltage pulse, shall not exceed the peak pulse value by more than 4,000 volts, and the duration of any spike (measured at a point 70 percent above the peak pulse value) shall not exceed 2 μs. In the absence of RF output voltage, the peak pulse supply voltage may be exceeded by 20 percent but by not more than 10 consecutive pulses. A pressurized output cavity is required, for operation at the maximum voltage specified herein, to prevent external flashover at the insulating bushing (see 9/).

3/ Peak positive screen voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portions of the pulse. The magnitude of any spike, on the grid(2) voltage pulse, shall not exceed the peak pulse value by more than 250 volts, and the duration of any spike (measured at a point 70 percent above the peak pulse value) shall not exceed 2 μs.

4/ On time is defined as the sum of the durations of all the individual pulses which occur during any 3,000 μs interval.

5/ Peak rectified grid(1) current.

6/ Minimum time to reach operating filament voltage shall be 30 seconds. After operating filament voltage has been obtained, a minimum of 90 seconds heating time is required before other electrode voltages can be applied. Filament current shall not exceed 550 amperes, even momentarily.

7/ Cooling requirements.

a. Air cooling. Forced-air cooling of the ceramic bushing at the grid(1) seal and at the anode seal may be required in order to limit the temperature of the ceramic bushing at either seal to the specified maximum value of +150°C. Under such conditions, provisions shall be made for blowing air at the ceramic bushing through suitable openings in the coaxial-cylinder cavity circuit.

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TABLE I. Testing and inspection - Continued.

b. Liquid cooling. Liquid cooling of the filament block, dc cathode block, grid(1) block, grid(2) block, and anode, is required. When the environmental temperature is above 0°C, the coolant may be water. The liquid flow shall start before application of any voltages and preferably shall continue for several seconds after removal of all voltages. Interlocking of the liquid flow through each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate liquid flow. Liquid flow shall be as shown in table IA.

TABLE IA. Liquid flow.

Coolant course	Waterflow		Maximum pressure differential 1/ for typical flow (psi)
	Absolute minimum (gpm)	Typical (gpm)	
With water:			
Through filament block - - - - - - - - - -	0.5	0.8	8
Through dc cathode block- - - - - - -	0.5	0.8	8
Through grid(1) block - - - - - - - - - - -	0.5	0.8	6
Through grid(2) block - - - - - - - - - - -	0.5	0.8	8
Through anode in direction shown on figure 1:			
For anode dissipation up to 11 kW (average) - - - - - - - - - -	8	10	10
For anode dissipation of 14 kW (average- - - - - - - - - -	10	12	15

1/ Directly across cooled element.

8/ The water flow to the anode cooling connection shall be 8 +1, -0 gpm. The minimum water flow to the filament, filament-cathode, grid(1) cooling connections, shall be 0.5 gpm. Water flow shall start before the application of any voltages, and shall be continued for at least 6 seconds after removal of all voltages. The outlet water temperature shall not exceed +70°C.

Emergency conditions: Simultaneous interruption of cooling pump and tube electrode power is permissible.

9/ Environmental requirements.

External gas pressure (absolute) 1/	60 psi (max) 2/
Gage pressure at any inlet	100 psi (max)
Metal-surface temperature	100°C max
Ceramic bushing temperature	150°C max
Outlet liquid temperature (any outlet) for water	70°C max
Storage temperature 3/	-65°C max

1/ Absolute pressure is the sum of the atmospheric pressure and gage pressure.
2/ This pressure is related to the output cavity pressurization as required to prevent corona or external flashover.
3/ The tube coolant ducts shall be free of liquid before storage or shipment of the tube to prevent damage from freezing.

10/ High-speed electronic protective devices shall be used to prevent energy from the anode power supply, from the grid(1) power supply, and from the grid(2) power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the various pulse sources supplying the tube during the inter-pulse interval following the occurrence of a fault in any single pulse, and shall be capable of meeting the requirements of the aluminum foil simulated fault tests. Thus, all pulses subsequent to a “fault” pulse shall not be applied until the device is reset. (A fault is a condition in which the peak pulse currents rise 20 percent or more above the normal peak current level.) Additional protection against damage from an anode-to-grid(2) fault shall be provided to prevent the grid(2) voltage from rising to a value in excess of 3,000 volts during the fault pulse. Minimum reset time shall be 1 second.

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TABLE I. Testing and inspection - Continued.

11/ This requirement shall be waived for manufacturers qualified to tube type 8932 (see 28/).

12/ Water system requirements. In order to minimize maintenance, it is recommended that the water cooling system be of the closed type, utilizing water of acceptable quality as specified herein, to prevent excessive deposition, corrosion, and electrolysis. Deposition of material not only restricts water flow, but also prevents proper transfer of heat from the tube elements to the cooling water. Corrosion and electrolysis may destroy the tube elements, ducts, and fittings. Cooling water of acceptable quality shall have a specific resistivity in excess of 1 megohm-cm.

13/ The water flow to the anode cooling connection shall be 10 gpm. The water flow to the filament, filament cathode, grid(1) and grid(2) cooling connections, shall be 0.8 gpm. Pressure drops are measured at the tube terminals.

14/ The tube shall operate continuously under the specified test conditions.

15/ The grid(1) voltage (Ec1) is adjusted for class B operation. Class B operation is defined as a negative Ec1 voltage which is between 20 percent and 12.5 percent of the ec2 voltage.

16/ Circuit adjustments are permissible. The rectified RF output pulse length (tp) and the pulse repetition rate (prr) shall be as specified herein. Peak output power is calculated by dividing the average output power by the duty factor. Duty factor is defined as the product of pulse duration and the repetition rate. The average output power is measured by a calorimetric method using a dummy water load, RCA Model MI 19198-A, or equivalent. Minimum output circuit bandwidth as checked by cold probe techniques shall be 12 MHz to the 3 dB points and 20 MHz to the 6 dB points.

17/ With the anode connected to the cathode and grid(2) connected to grid(1), apply a partial sinusoidal voltage of 200 v peak to the grid(1). Emission current shall be read on an oscilloscope connected across a noninductive resistor in series with the grid(1)-grid(2) cathode circuit or by means of a current transformer.

18/ The RF input power is measured using a calibrated directional coupler. This power is the difference between the incident and the reflected power as sampled by the directional coupler. Coupler calibration is accomplished by the calorimetric method, RF substitution method, or other approved methods. Power gain is calculated by dividing the RF output power by the RF input power. The grid circuit shall be adjusted for lowest VSWR in the drive line. Specification readings cannot be taken unless reflected power is less than 10 percent of the incident power.

19/ With conditions as specified herein, grid(1) current (I_1) shall be recorded after specified time. All voltages shall be removed except filament voltage and grid(1) voltage, and then grid(1) leakage current (I_2) shall be recorded. Calculate Ic from the formula: $Ic = I_1 - I_2$.

20/ The following procedure (or equivalent) shall be used for measuring input and output resonant frequency.

a. Set up equipment as in figure 2. Be sure tube is making good contact with cavity.

b. Adjust the frequency of RF generator to obtain a resonant peak on the standing-wave indicator.

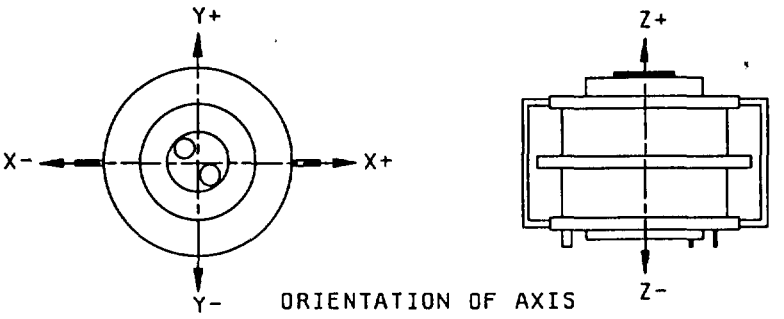
c. The resonant frequency as read on frequency meter shall be within limits specified herein. (The input cavity and output cavity shall be as shown on figures 3 and 4.)

21/ Each element not under test shall be tied to ground. Two tubes per month shall be tested. Failure of one or more tubes will require 100-percent testing. Two tubes per month may be resumed after testing of 20 consecutive tubes without failure.

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TABLE I. Testing and inspection - Continued.

22/ Vibration and shock tests shall be performed using vibration-shock fixture (DSCC Drawing No. 67031), or equivalent. Tests shall be performed with no voltages applied.



ORIENTATION OF AXIS

Post vibration and shock need only be performed once (at the conclusion of both vibration and shock tests).

23/ Vibration.

a. The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)
5 to 55	0.060

b. Frequency shall be swept logarithmically from 5 to 55 to 5 Hz in 5 minutes minimum. Two sweeps shall be made.

c. The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)	Duration (hours/axis)
50	0.060	12

d. Tubes which do not comply with the post-vibration limits shall be considered failures.

24/ Shock. Test in accordance with MIL-STD-202, method 213, test condition A, except as follows:

a. The peak acceleration of the shock shall be 15 G ± 10 percent.

b. The velocity change (V_i) = 3.39 feet/second.

c. Tubes which do not comply with post-shock limits shall be considered failures.

d. This is not a destructive test.

25/ This test is to be the first test performed at the conclusion of the holding period.

26/ The anode current pulse shall be examined for evidence of pulse droop. No pulse droop attributable to tube emission shall be permitted.

27/ The tube shall operate continuously under power output conditions for 1 hour without a trip-out. Measured data to be taken at the end of this 1-hour period. A trip-out is defined as an increase in plate current in excess of 20 percent above normal level.

28/ Tube type 8932 shall be used to replace tube types 4666 and 6952 in AN/SPS-40A, AN/SPS-40B, AN/SPS-40C, and AN/SPS-40D applications. Tube type 6952 shall be used in AN/SPS-29, AN/SPS-32, and AN/SPS-40 applications. Users should evaluate tube type 8932 before using it as a replacement in other applications.

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TABLE I. Testing and inspection - Continued.

29/ Emission check. Test conditions and limits for this test shall be as specified. However, the following alternate method may be used:

Test conditions	Test limits
Ef/If = 450 A, 60 Hz; ebb = 30 kV; ec2 = 2 kv (max); ec1 = 15/; RF drive/Ib = 165 mA dc (max) 16/; Ic1 = 0 to +28 mA dc; F = 225 ± 15 MHz; tp = 10 ± 1 µs; prr = 300 ± 5	po = 825 kW minimum

30/ The tube shall have passed the Ic1 limit at the end of 1 hour if Ic1 ≥ 15 mA dc. If, at the end of 1 hour, the Ic1 reading is between 10 to 14 mA dc inclusive, then continue the test for an additional hour after which Ic1 shall equal or exceed 10 mA dc.

31/ The following life test may be used as an alternate to initial life at the option of the manufacturer.

Test	Notes	Conditions	Symbol	Minimum	Maximum	Unit

Initial life (alternate)	1, 11, 12, 15, 16	Ef = 0.95 V, 60 Hz;
		Ebb = 22 kv dc (max);
		ec2 = 2 kv (max);
		Ec1 = 15/;
		pd/Ib = 0.50 Adc;
		po = 200 kw (min);
		F = 200 MHz (min);
		tp = 100 µs (min);
		prr = 240 ± 5;
		t = 1,000 hours
			---	---	---	---

Initial life (alternate) test end point:						
Peak power output	---	Initial life (alternate)	po	200	---	kw

32/ The following life test may be used as an alternate to multiple short life at the option of the manufacturer:

Test	Notes	Conditions	Symbol	Minimum	Maximum	Unit
Multiple short life (alternate)		Same as initial life (alternate, 31/) except, po = 225 kw (min); t = 50 hours	---	---	---	---
Multiple short life (alternate) test end point:						
Peak power output	---	Multiple short life (alternate)	po	225	---	kw

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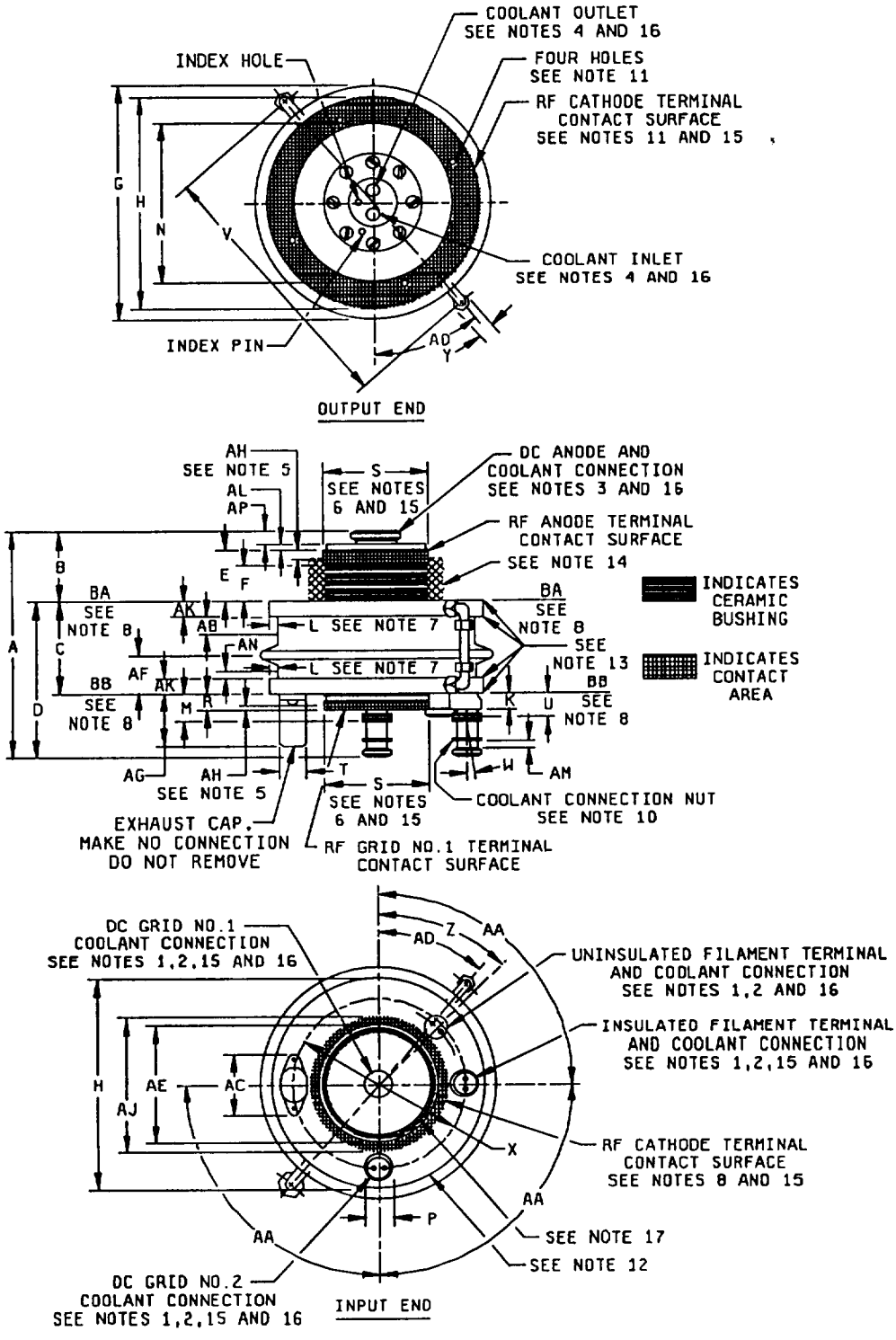


FIGURE 1. Outline drawing of electron tube type 8932.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
Conformance inspection, part 1					16
	G-3 gage check				15
E	1.83	1.93	46.48	49.02	
R	.59	.69	14.99	17.53	
S	3.74	3.76	95.00	95.50	6, 15, 18
Conformance inspection (periodic check) part 3					
A	8.31	8.93	211.07	226.82	
C	3.27	3.45	83.06	87.63	
D	5.68	6.18	144.27	156.97	
F	1.17	---	29.72	---	
G	---	8.75	---	222.25	
H	7.84	7.88	199.13	200.15	18
K	---	.59	---	14.99	
L	.13	---	3.30	---	7, 18
M	1.04	1.14	26.42	28.96	15
P	1.22	1.28	30.99	32.51	
T	.97	1.12	24.64	28.45	
U	.83	.95	21.08	24.13	
V	---	11.25	---	285.75	
Y	---	.68	---	17.27	
AB	.50	---	12.70	---	7
AC	2.09	2.15	53.09	54.61	
AE	---	4.06	---	103.12	9
AF	1.38	1.50	35.05	38.11	
AG	---	2.00	---	50.80	
AH	.38	---	9.65	---	5
AJ	4.95	---	125.73	---	9
AK	.56	.63	14.22	16.00	
AL	---	.28	---	7.11	
AM	.09	---	2.29	---	
AN	.25	---	6.35	---	7
AP	.52	.60	13.21	15.24	
Reference dimensions					
B	---	2.69	---	68.33	15
J	---	3.25 RAD	---	82.55 RAD	15
N	---	6.00	---	152.40	11
W	3°		3°		15
X	---	6.25	---	158.75	15
Z	45°		45°		15
AA	90°		90°		15
AD	41°		41°		15

FIGURE 1. Outline drawing of electron tube type 8932 - Continued.

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NOTES:

1. Terminal has 1 inch (25.4 mm) -16 unified thread, class 2A fit, 0.380 inch (9.65 mm) long, and two holes 0.258 inch (6.55 mm) to 0.270 inch (6.86 mm) diameter, spaced 0.438 inch (11.13 mm) on centers.
2. The holes in the filament, grid(1), and grid(2) coolant terminal connections, will accept the pins of the plug-and-cylinder combination gage G1, shown on figure 5.
3. Terminal has 1-3/4 inch (44.45 mm)-16 unified extra fine thread, class 2A fit, 0.380 inch (9.65 mm) long, two holes 0.508 inch (12.90 mm) to 0.522 inch (13.26 mm) diameter, spaced 0.688 inch (17.48 mm) on centers, and an index hole 0.160 inch (4.06 mm) maximum diameter, spaced 0.344 inch (8.74 mm) from the center of the terminal.
4. Holes in the anode coolant connections shall accept the pins of the plug-and-cylinder combination gage G2, shown on figure 6.
5. Pressure from circuit contacts shall be exerted only over maximum length, AH, of designated contact areas of the anode or grid(1) terminals.
6. This diameter dimension is held only over a length of AH.
7. Dimension L applies over length AB and AN, as indicated.
8. The contact surfaces, BA-BA and BB-BB of the RF cathode terminals, shall be parallel within 0.060 inch (1.52 mm).
9. Contact of the input end, RF cathode terminal, shall not be made at a diameter smaller than 4.060 inch (103.12 mm), nor greater than 4.950 inch (125.73 mm).
10. To prevent excessive stress on the ceramic seal, a 15/16 inch (23.81 mm) open-end wrench shall be used to permit gripping the terminal when removing or tightening the coolant connectors.
11. Contact of the output end, RF cathode terminal, shall not be made at a diameter smaller than 6.000 inches (152.40 mm). The pressure exerted for this RF contact shall be limited to that necessary for good electrical contact. The mechanical force for the cavity support and pressure seal shall be made at a diameter not less than 6.000 inches (152.40 mm). On the output end, RF cathode terminal, there are four equally spaced 0.188 inch (4.78 mm) diameter holes on a circle having a diameter of 6.750 inches (171.45 mm). These holes are for tube manufacturing purposes only. Attention is called to the existence of these holes so that equipment designers can avoid making a pressure seal or electrical contact at points which are coincident with these holes. Mechanical clamping devices for the output cavity shall be designed so as to exert their clamping force across outer edge of output-header flange.
12. Serial number is located on this surface between grid(2) and the insulated filament terminal.
13. Corners may be rounded or chamfered, but not to exceed .050 inch (1.27 mm).
14. Keep this 5-inch diameter annular-volume region clear. Do not allow circuit components (conductors or insulators) to protrude into this region.
15. The output end shall accept the output end of gage G3, shown on figure 7. Proper acceptance is obtained when gage is seated on the output end, RF cathode terminal, contact surface. The gage is properly seated when a 0.015 inch (0.38 mm) thickness gage, 1/2 inch (12.70 mm) wide will not enter between the gage and the RF cathode contact surface. The pins on the 1.500 inch (38.10 mm) movable section of the output end of gage G3, shall enter the anode coolant holes such that the height slot represented by the 2.098 inch (53.29 mm) dimensions is below the shoulder of the gage, and the 2.219 inch (56.36 mm) dimension is visible above the shoulder. The input end shall accept the input end of gage G3, shown on figure 7. Proper acceptance is obtained when the gage is seated on the input end, RF cathode terminal surface, and the projecting guide pins of the output end engage the mating holes of the input end. The gage is properly seated when a 0.015 inch (0.38 mm) thickness gage, 1/2 inch (12.70 mm) wide will not enter between the gage and the RF cathode terminal contact surface. The pins on the 0.875 inch (22.23 mm) movable section of the input end shall enter the grid coolant holes such that the height slot represented by the 2.379 inch (60.43 mm) dimension is below the shoulder of the gage, and the 2.559 inch (65.00 mm) dimension is visible above the shoulder, when gage G3 is properly seated.
    a. Dimensions B, J, M, W, X. Z, AA, and AD shall vary only within the limits of the gage.
    b. The centerline through the grid(1) coolant connection holes will have a maximum angle of 2 degrees with respect to the centerline of the anode coolant connection holes.

FIGURE 1. Outline drawing of electron tube type 8932 - Continued.

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c. The axis of the RF anode terminal contact surface is coincident with the axis of the grid(1) terminal contact surface, within 0.050 inch (1.27 mm).

d. The axis of the anode and coolant connection is coincident with the axis of the RF anode terminal contact surface, within 0.025 inch (0.64 mm).

16. Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0, inspection level II.

17. The outside diameter of this ceramic bushing shall not enter the annular region defined by a circle of 3.880 inch (98.55 mm) diameter concentric with diameter S of the RF grid(1) terminal.

18. Measurement of this dimension shall be made on both input and output ends.

19. The centers of the water course holes, in each connection around the periphery of the tube, shall be on a line through the center of these holes and perpendicular to a line through the center of the tube within ±2 degrees.

FIGURE 1. Outline drawing of electron tube type 8932 - Continued.

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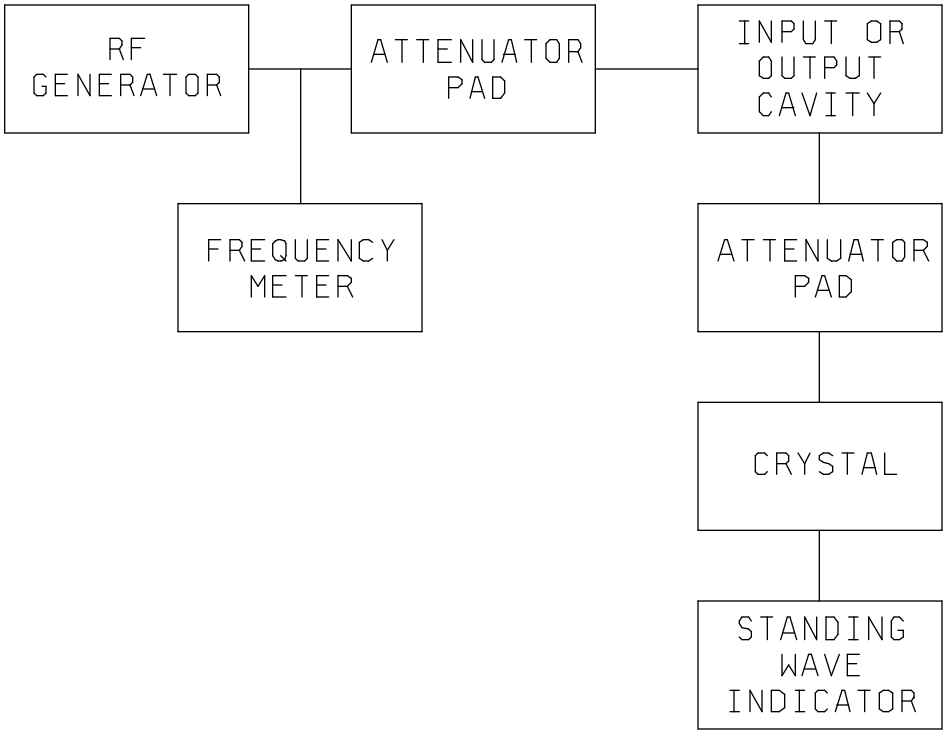
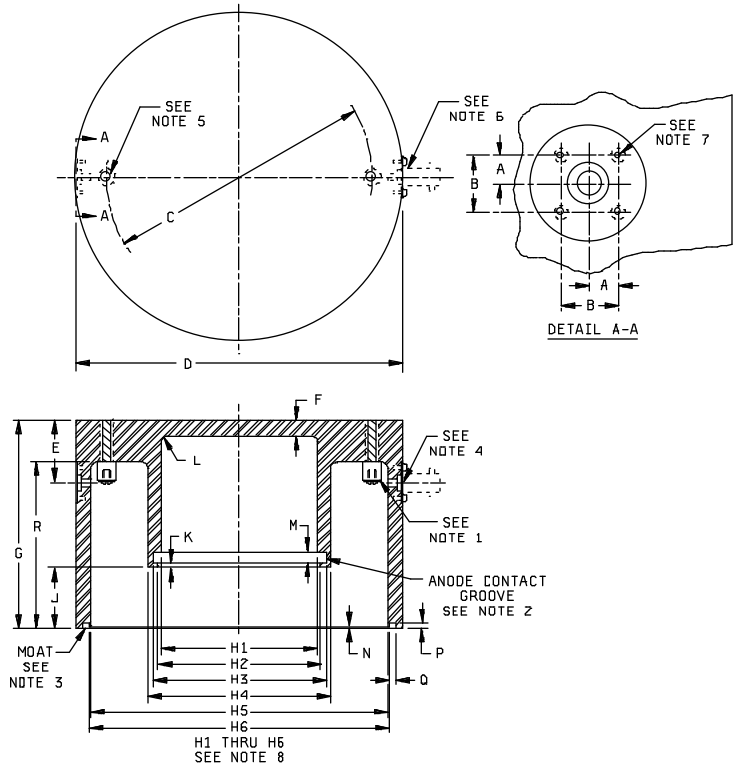


FIGURE 2. Typical block diagram of test circuit.

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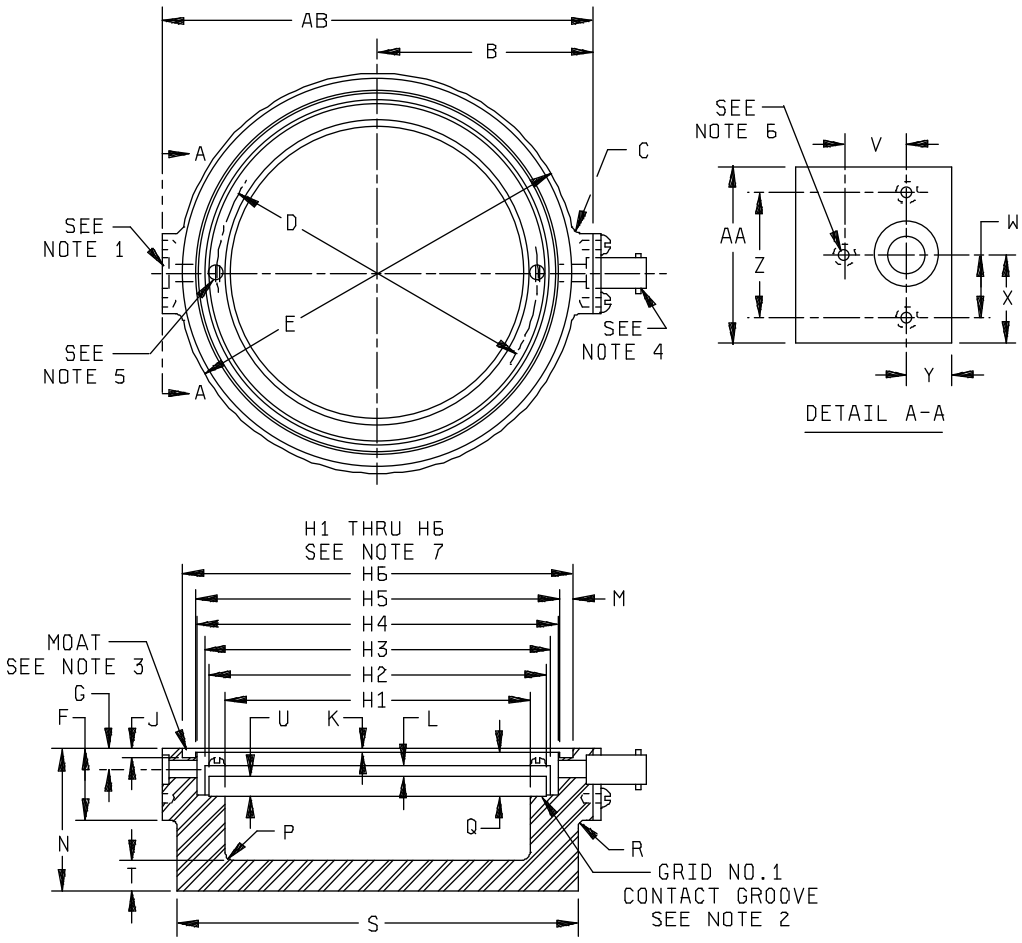
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.250 REF		6.35 REF	
B	.490	.510	12.45	12.95
C	6.397 DIA	6.403 DIA	162.48 DIA	162.64 DIA
D	7.885	7.905	200.28	200.79
E	1.485	1.515	37.72	38.48
F	.365	.385	9.27	9.78
G	4.985	5.015	126.62	127.38
H1	3.770	3.780	95.76	96.01
H2	3.805	3.811	96.65	96.80
H3	4.169	4.175	105.89	106.05
H4	4.401	4.403	111.79	111.84
H5	7.145	7.147	181.48	181.53
H6	7.243	7.249	183.97	184.12
J	1.466	1.468	37.24	37.29
K	.098	.104	2.49	2.64
L	.12 R TYP		3.05 R TYP	
M	.253	.259	6.43	6.58
N	.009	.015	0.23	0.38
P	.107	.113	2.72	2.87
Q	.153	.159	3.89	4.04
R	3.995	3.997	101.47	101.52

NOTES:

1. Two studs. 1-½-inches (38.10 mm) long threaded ¼-inch (6.35 mm)-20 NC by 1-inch (25.40 mm) with 3/8-inch (9.53 mm) diameter head. Head of stud shall be drilled and tapped ¼-inch (6.35 mm) deep for No. 6-32 RH machine screw. Use standard 5/32-inch (3.97 mm) flat washer under screw.
2. Anode contact groove shall be filled with the following spring: 0.020-inch (0.51 mm) diameter wire wound to give a 0.250-inch (6.35 mm) O.D. at 35 TPI, 14-¼-inches (361.95 mm) in length (approximate).
3. Moat shall be filed with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the RF cathode terminal contact surface.
4. Two holes. Drill 7/32-inch (5.56 mm) diameter (through), C' bore 3/8-inch (9.53 mm) diameter by 1/8-inch (3.18 mm) deep. Spot face 1-inch (25.40 mm) diameter by 1/16-inch (1.59 mm) deep.
5. Two holes. Drill and tap for ¼-inch (6.35 mm)-20 NC screws.
6. Location of A-N connector - UG-290/U (two connectors 180 degrees apart).
7. Four holes. Drill and tap for No. 3-48 screws by ¼-inch (6.35 mm) deep.
8. Cylindrical holes H1 through H6 have axes coincident within 0.001 inch (0.025 mm).

FIGURE 3. Output cavity required for resonant frequency test.

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NOTES:

1. Two holes, drill 7/32-inch (5.56 mm) diameter (through), C' bore 3/8-inch (9.53 mm) diameter by 1/16-inch (1.588 mm) deep.
2. Grid(1) contact groove to be filled with the following spring: 0.020-inch (0.51 mm) diameter wire wound to give a 0.250-inch (6.35 mm) O.D. at 35 TPI, 14-¼-inches (361.95 mm) in length (approximate).
3. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the RF cathode terminal contact surface.
4. Location of A-N connector - UG-290/U (two connectors 180 degrees apart).
5. Two holes. Drill and tap for No. 3-48 screws.
6. Three holes. Drill and tap for No. 3-48 screws by 3/16-inch (4.78 mm) deep.
7. Cylindrical holes H_1 through H_6 have axes coincident within 0.001 inch (0.025 mm).

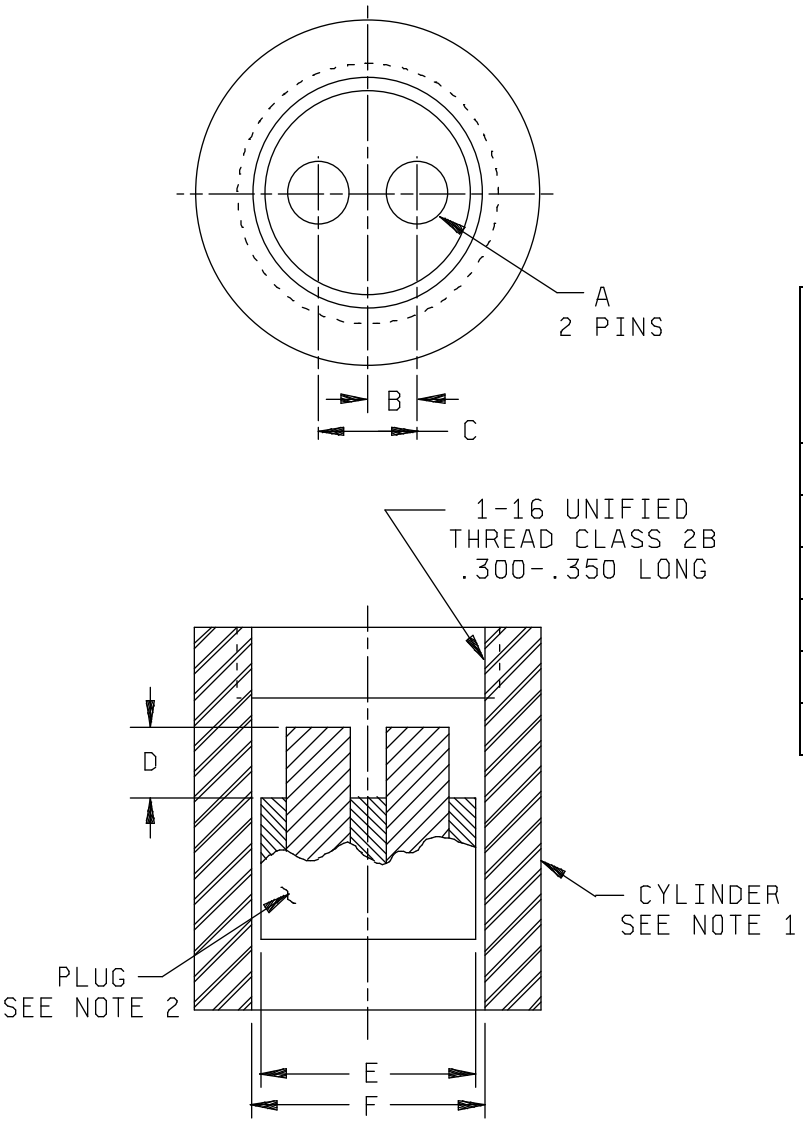
FIGURE 4. Input cavity required for resonant frequency test.

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Ltr	Dimensions				Ltr	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Max	Max
B	2.657	2.667	67.49	67.74	M	.153	.159	3.89	4.04
C	.12 R TYP		3.05 R TYP		N	1.740	1.760	44.20	44.70
D	3.987	3.993	101.27	101.42	P	.12 R		3.05 R	
E	4.947	4.953	125.65	125.81	Q	.574	.576	14.58	14.63
F	.865	.885	21.97	22.48	R	.12 R		3.05 R	
G	.244	.254	6.20	6.45	S	4.947	4.953	125.65	125.81
H1	3.765	3.775	95.63	95.89	T	.365	.385	9.27	9.78
H2	3.802	3.808	96.57	96.72	U	.252	.258	6.40	6.55
H3	4.169	4.175	105.89	106.05	V	.351	.357	8.92	9.07
H4	4.273	4.275	108.53	108.59	W	.354 REF		8.99 REF	
H5	4.450	4.452	113.03	113.08	X	.500 REF		12.70 REF	
H6	4.505	4.511	114.43	114.58	Y	.247	.253	6.27	6.43
J	.105	.111	2.67	2.82	Z	.702	.712	17.83	18.08
K	.012	.018	0.30	0.46	AA	.990	1.010	25.15	25.65
L	.125	---	3.18	---	AB	5.315	5.335	135.00	135.51

FIGURE 4. Input cavity required for resonant frequency test - Continued.

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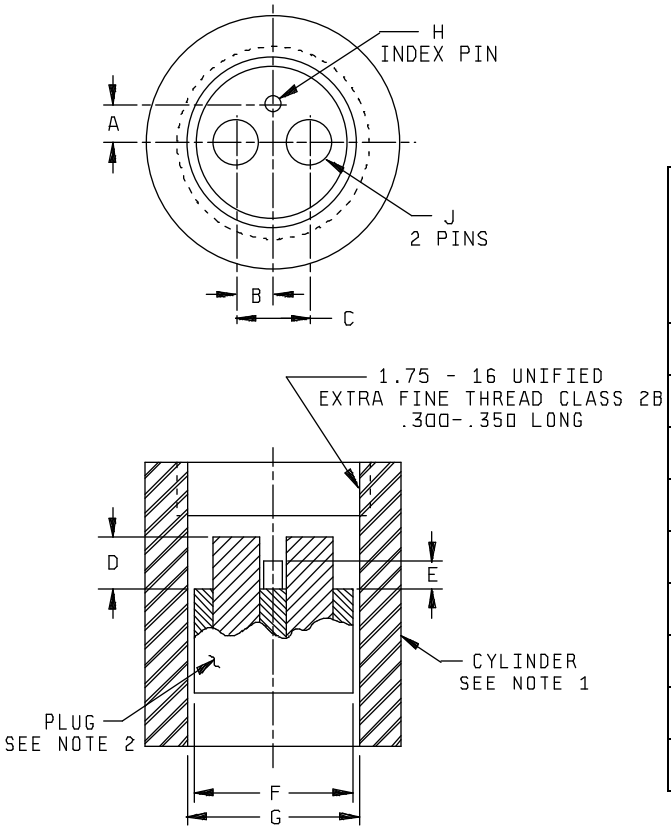
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.255	.257	6.48	6.53
B	.219 REF		5.56 REF	
C	,436	.440	11.07	11.18
D	.178	.198	4.52	5.03
E	.891	.899	22.63	22.83
F	.906	.914	23.01	23.22

NOTES:

1. Axis of the cylindrical holes and threaded section shall be coincident within 0.002 inch (0.05 mm).
2. Plug sides and pin sides shall be parallel within 0.001 inch (0.025 mm).
3. Dimensions are in inches. Tolerances shown are gage tolerances and shall not be considered tube tolerances.

FIGURE 5. Gage G-1.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.341	.347	8.66	8.81
B	.344 REF		8.74 REF	
C	.686	.690	17.42	17.53
D	.490	.510	12.45	12.95
E	.240	.260	6.10	6.60
F	1.621	1.629	41.17	41.38
G	1.656	1.664	42.06	42.27
H	.120 DIA	.130 DIA	3.05 DIA	3.30 DIA
J	.505	.507	12.83	12.8

NOTES:

1. Axis of the cylinder holes and threaded section shall be coincident within 0.002 inch (0.05 mm).
2. Plug sides and pin sides shall be parallel within 0.001 inch (0.025 mm).
3. Dimensions are in inches. Tolerances are gage tolerances and shall not be considered tube tolerances.

FIGURE 6. Gage G-2.

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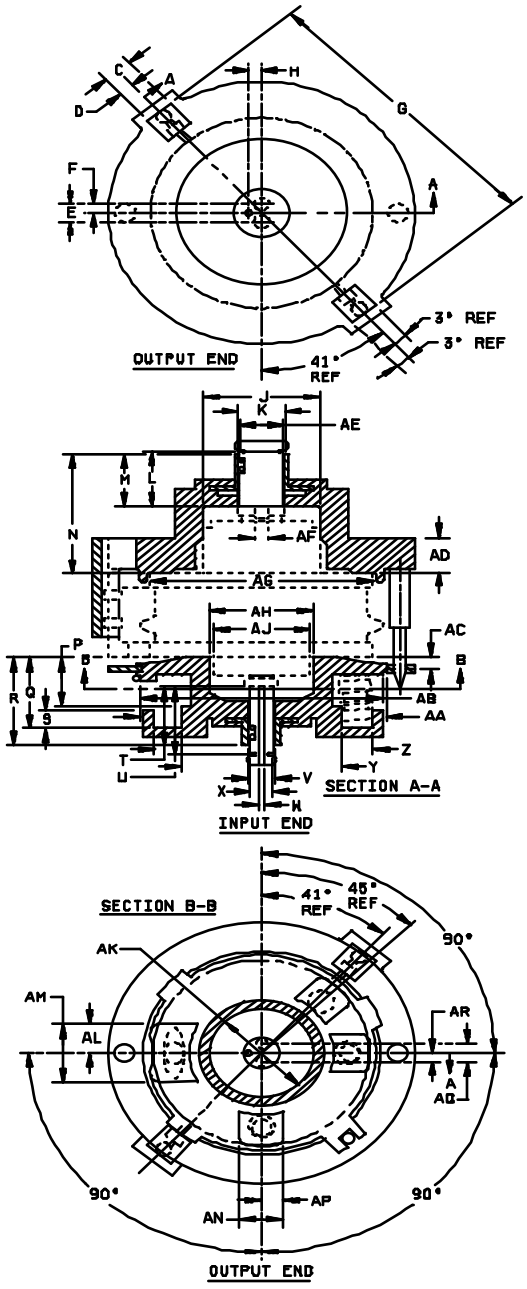


FIGURE 7. Gage G-3.

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Ltr	Dimensions				Ltr	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
C	1.050	1.058	26.67	26.87	X	.875 DIA		22.23 DIA	
D	.527 (REF)		13.39 (REF)		Y	5.378	5.388	136.60	136.86
E	.686	.690	17.42	17.53	Z	7.731	7.741	196.37	196.62
F	---	.344	---	8.74	AA	8.337 (REF) DIA		211.76 (REF) DIA	
G	11.235	11.255	285.37	285.88	AB	8.437 (REF) DIA		214.30 (REF) DIA	
H	.344 (REF)		8.74 (REF)		AC	.215	.221	5.46	5.61
J	3.759	3.763	95.48	95.58	AD	1.563	1.573	39.70	39.95
K	1.539 (REF)		39.09 (REF)		AE	1.500 (REF)		38.10 (REF)	
L	2.214	2.224	56.24	56.49	AF	.505	.506	12.83	12.85
M	2.093	2.103	53.16	53.42	AG	5.995	6.005	152.27	152.53
N	4.848	4.858	123.14	123.39	AH	4.058	4.062	103.07	103.17
P	1.981	1.991	50.32	50.57	AJ	3.759	3.763	95.48	95.58
Q	2.839	2.849	72.11	72.36	AK	4.947 DIA	4.953 DIA	125.65 DIA	125.81 DIA
R	3.573	3.583	90.75	91.01	AL	---	1.250	---	31.75
S	.682	.692	17.32	17.58	AM	2.490	2.510	63.25	63.75
T	2.374	2.384	60.30	60.55	AN	1.615 (TYP)	1.635 (TYP)	41.02 (TYP)	41.53 (TYP)
U	2.554	2.564	64.87	65.13	AP	.812 (TYP)		20.62 (TYP)	
V	.915 DIA		23.24 DIA		AQ	.436	.440	11.07	11.18
W	.255	.257	6.48	6.53	AR	---	.219	---	5.56

NOTES:
1. Tolerances shown are gage tolerances and shall not be considered tube tolerances.

2. See DSCC Drawing 67038.

FIGURE 7. Gage G-3 - Continued..

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Review activities:
Navy - AS, CG, MC, OS, SH

Preparing activity:
DLA - CC

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